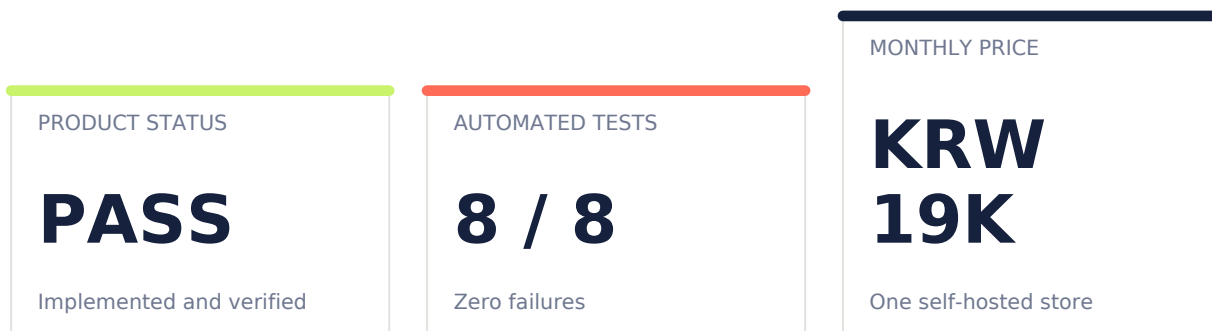


SYSTEM FOUNDRY / BUSINESS MVP

ClaimClock

A deadline-first operations desk for returns, exchanges, and refunds.



01 / PROJECT OVERVIEW

Project overview

ClaimClock is a local-first operations system that turns scattered return, exchange, and refund requests into a prioritized work queue. It was built as a single-admin MVP with no paid API dependency.

02 / TARGET CUSTOMER

Target customer

Small Korean online merchants processing five or more customer claims per day, especially Smart Store and direct-to-consumer operators still coordinating work through chat messages and spreadsheets.

03 / PROBLEM

Problem

Claim details arrive through several channels. A spreadsheet records the case but rarely surfaces urgency, so due dates slip, customer trust declines, and the owner cannot see how much money remains exposed.

PRODUCT

A narrow workflow with three jobs

04 / CORE FEATURES

Core features

The MVP deliberately limits scope to the following operational loop:

01	Validated claim intake Order reference, case type, amount, deadline, alias, and note with server-side limits.
02	Deadline risk dashboard Overdue, urgent (48 hours), watch, safe, and completed classifications calculated from the current date.
03	Status and exposure tracking Open, waiting, and resolved transitions with active-case count and financial exposure total.

05 / HOW TO USE

How to use

Set two required environment variables, start the Python server, sign in with the configured access code, register a case, and work from the highest-risk row. Change the status to waiting or resolved as processing advances.

06 / SYSTEM ARCHITECTURE

System architecture

The server is intentionally compact and auditable:

Browser	HTTP handler	Domain rules	SQLite
Server-rendered UI	Session + CSRF	Validation + risk	Parameterized SQL

07 / TECHNOLOGY

Technology

Python 3 standard library, SQLite, server-rendered HTML/CSS, and unittest. Runtime dependencies: zero. External paid APIs: zero.

VALIDATED INTERFACE

Actual running screen

Captured from headless Chromium after login and four real form submissions against a temporary SQLite database.

The screenshot displays the ClaimClock dashboard. At the top, there's a header with the ClaimClock logo and a '로그아웃' (Logout) button. A green notification bar states '변경사항을 저장했습니다.' (Saved changes). Below this, a section titled 'TODAY'S CONTROL TOWER' features a large Korean text overlay: '놓치기 전에, 먼저 처리하세요.' (Before you miss it, process it first). A date badge shows '2026.08.02'.

Three key metrics are shown in colored boxes:

- 진행 중 4** (Active Cases): 4 ACTIVE CASES
- 긴급 처리 2** (Urgent Cases): 2 DUE IN 48H
- 처리 대상 금액 ₩271,800** (Open Exposure): ₩271,800 OPEN EXPOSURE

The main content area is divided into two panels:

- CASE QUEUE 처리 현황** (Case Queue Processing Status): A table with 4 cases.

위험도	주문번호	유형	금액	기한	상태	행동
긴급	ORD-260802-41 ..	환불	₩78,400	오늘 2026-08-02	접수	접수 ▼ 저장
긴급	ORD-260802-38 이**	교환	₩42,900	D-2 2026-08-04	접수	접수 ▼ 저장
주의	ORD-260801-19 ..	반품	₩119,000	D-4 2026-08-06	접수	접수 ▼ 저장
안전	ORD-260731-07 장**	환불	₩31,500	D-7 2026-08-09	접수	접수 ▼ 저장
- NEW CASE 처리 건 등록** (New Case Processing Registration): A form to add a new case.
 - 주문번호: ORD-2026-0182
 - 고객 별칭: 김**
 - 유형: 반품 (Amount: 0)
 - 처리기한: 08/05/2026
 - 메모: 접수 후 환불 예정
 - Button: 처리 건 추가

08 / RUNNING SCREEN

Running screen

The captured dashboard shows four active cases, two due within 48 hours, and KRW 271,800 in open exposure. Korean typography is bundled locally, so the interface does not depend on a third-party font CDN.

VERIFICATION

Evidence, not assumptions

09 / TEST RESULTS

Test results

Every required check was executed against the generated project.

Check	Result	Evidence
Dependency installation	PASS	Fresh venv + empty runtime requirements
Application start	PASS	GET /health returned HTTP 200
Core workflow	PASS	Login + four creates + dashboard totals
Automated tests	8 / 8	Validation, risk, database, status, duplicates
README commands	PASS	Install, test, and start commands executed
Credential scan	PASS	No secret patterns or .env file
PDF	PASS	Generated, parsed, rendered, visually inspected

10 / SECURITY AND LIMITATIONS

Security and limitations

No embedded credentials or default admin account. Access key and HMAC session secret are required environment variables. Mutations require an authenticated HttpOnly SameSite cookie and CSRF token. Inputs are length-, type-, amount-, and date-validated; SQL uses parameter binding. The default bind address is localhost. Limitations: single administrator, no TLS termination, no team permissions, no attachment malware scanning, and no marketplace synchronization.

GO TO MARKET

A sellable wedge

11 / PRICING AND REVENUE MODEL

Pricing and revenue model

KRW 19,000 per month for one self-hosted store, including updates and email support. A later hosted tier can charge KRW 39,000 per month for automated backups and team access. Ten pilot stores at the base tier produce KRW 190,000 MRR before hosting costs.

12 / FIRST CUSTOMER ACQUISITION

First customer acquisition

Recruit five operators from Korean Smart Store seller communities. Offer a 14-day pilot in exchange for a 20-minute workflow interview and permission to measure missed-deadline rate. Demonstrate the product with the seller's anonymized last ten claim cases. Convert with a simple promise: one board, every due date visible.

13 / IMPROVEMENT PLAN

Improvement plan

Prioritized follow-up work after real usage:

- CSV import/export and daily encrypted backup.
- Team accounts with role-based permissions and an audit timeline.
- Email or messenger reminders without exposing customer information.
- Optional marketplace connectors after validating demand and terms.

COMMERCIAL ASSUMPTION

The strongest initial sales angle is operational risk reduction, not generic CRM. ClaimClock should stay specialized until interviews show that merchants will pay for broader support tooling.

ASSESSMENT

Overall evaluation

14 / OVERALL SCORE

Overall score

Six dimensions are scored out of 10. Higher maintenance score means easier maintenance.

Dimension	Score	Rationale
Customer problem strength	8/10	Missed claim deadlines create direct refund cost and trust damage.
One-day completeness	9/10	The full intake-to-resolution loop is implemented and browser-tested.
Sales potential	7/10	Clear niche and price, but interviews must confirm willingness to switch.
Maintenance difficulty	8/10	Zero runtime dependencies and a compact SQLite design reduce upkeep.
Differentiation	7/10	Deadline-first positioning is sharper than spreadsheets, though replicable.
Scalability	8/10	Team roles, reminders, and connectors create credible expansion paths.

OVERALL

7.8 / 10

Proceed to five-customer pilot. Keep scope focused on deadline visibility.

Verdict: ClaimClock is technically complete as an MVP and commercially specific enough for pilot sales. The next evidence should come from merchant interviews, not additional features.